

NOIRÉA: Session-Bounded Identity Isolation in Adaptive Physical AI Systems

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Abstract—Physical AI systems increasingly operate in sustained interaction with human users across multiple deployment sessions, adapting their internal models based on human-derived signals. While this adaptive capability improves task performance, it introduces a previously unaddressed structural risk: the unintended accumulation of persistent, identity-relevant representations across session boundaries. We formalize this phenomenon as Persistent Identity Modeling (PIM), a structural risk class arising from unconstrained cross-session adaptive state persistence. Existing safety and governance frameworks address malfunction, privacy breaches, and behavioral misalignment, but do not specify architectural requirements for representation lifecycle management in adaptive systems. This paper introduces NOIRÉA, a structural constraint architecture that addresses PIM through four domain-agnostic architectural requirements enforcing session-bounded isolation of identity-correlated model state. We provide empirical evidence of observability through a minimal synthetic experiment showing measurable identity accumulation under persistent adaptation, and illustrate architectural instantiation through an adaptive Brain-Computer Interface case study. By characterizing cross-session representation persistence as a distinct reliability-relevant risk class, this work contributes to dependable system design by establishing representation lifecycle integrity as an architectural concern.

Index Terms—Adaptive systems, Physical AI, architectural constraints, representation lifecycle, session boundaries, identity isolation, dependable systems, human-AI interaction

I. INTRODUCTION

A. The Structural Risk of Adaptive Physical AI

Physical AI systems—encompassing robotics, assistive devices, neural interfaces, and sensor-driven embodied agents—are increasingly deployed in direct, sustained interaction with human users. Unlike conventional software systems, these systems acquire physical presence: they sense, model, and act within environments that include human bodies, movements, and physiological signals.

A defining characteristic of modern Physical AI is continuous adaptation. Systems iteratively update internal representations based on observed human interaction signals, improving task performance, reducing latency, and anticipating user intent. In clinical and assistive contexts, this adaptation is a primary feature: adaptive prosthetics recalibrate to user motor patterns; brain-computer interfaces refine neural decoders across sessions; collaborative robots update task models based on human behavioral feedback.

This adaptive capacity, however, introduces a structural risk that existing frameworks are not designed to address: the unintended accumulation of persistent, identity-relevant representations derived from human interaction signals across deployment sessions.

We refer to this risk class as **Persistent Identity Modeling (PIM)**.

B. Persistent Identity Modeling as a Structural Risk Class

PIM arises from standard adaptive learning procedures operating without architectural constraints on representation persistence. It does not require adversarial intent, data exfiltration, or system malfunction.

Over successive interaction sessions, model state may accumulate identity-correlated structure: feature clusters that distinguish one user from another, behavioral signatures that remain consistent across time, or latent embeddings that implicitly encode user identity without explicit design intent.

The structural conditions enabling PIM are common to a broad class of Physical AI architectures:

- Adaptive model updates occur across sessions without explicit state isolation
- Feature representations are reused across sessions without identity-decorrelation
- Session boundaries are not enforced at the representation layer
- No explicit reset mechanism ensures identity-uncorrelated model state at session transition

PIM is treated here as a structural risk class rather than a runtime failure. It emerges from a specification gap: current safety and governance frameworks do not impose architectural constraints on cross-session representation persistence. As a result, identity-correlated state accumulation may occur by default under standard adaptive optimization procedures.

C. Limitations of Existing Approaches

Current mechanisms for managing human-AI interaction in Physical AI systems are not designed to provide structural mitigation of PIM.

Informed consent frameworks establish user agreement prior to deployment but cannot anticipate the emergence of identity-correlated representations as a byproduct of adaptive optimization over extended deployment periods. Consent frameworks operate at the policy layer; PIM operates at the architectural layer.

Data minimization and privacy-by-design principles constrain data collection and retention but do not specify architectural requirements for representation lifecycle management within adaptive model pipelines. A system may collect minimal raw data while still accumulating persistent identity embeddings through iterative model adaptation.

Differential privacy and federated learning [3]–[5] provide statistical guarantees against specific inference attacks but do not prevent the formation of stable identity-correlated feature subspaces within session-local model parameters.

Functional safety standards commonly applied to Physical AI systems (e.g., IEC 61508 [1], ISO 26262 [2]) address malfunction and loss of control, but do not specify requirements for representation lifecycle management within adaptive pipelines.

Corrigibility and shutdown mechanisms ensure that systems respond to explicit human commands. They do not constrain the internal representation dynamics of adaptive models between commanded interactions.

The gap is architectural: no existing framework specifies structural requirements for preventing cross-session identity accumulation within Physical AI adaptive pipelines.

D. Contribution: NOIRÉA

This paper introduces **NOIRÉA**, a structural constraint architecture that enforces session-bounded identity isolation in adaptive Physical AI systems.

NOIRÉA specifies four architectural requirements:

- **R1 — Session-Bounded Identity Isolation:** At session initialization, identity-correlated model state must be reset to a zero or baseline configuration, preventing cross-session state carryover by default.
- **R2 — Controlled Cross-Session Persistence:** Any persistence of identity-correlated state across session boundaries must be explicitly user-authorized, logically separable from task-adaptive state, and revocable without affecting task performance.
- **R3 — Bounded Identity Influence:** The measurable influence of identity-correlated model state on system behavior must remain below a predefined detectability threshold.
- **R4 — Revocable Identity State Reset:** Users must be able to revoke identity-correlated state independently of task-adaptive state through a verifiable reset mechanism.

These requirements are designed to be:

- **Domain-agnostic:** applicable across Physical AI system classes without prescribing specific implementation techniques
- **Composable:** integrable into existing adaptive learning pipelines without full architectural replacement
- **Verifiable:** each requirement admits formal compliance checking against system architecture and dataflow properties

NOIRÉA does not prohibit adaptation. It constrains representation persistence to prevent unintended cross-session identity accumulation.

E. Validation Approach

We validate the NOIRÉA requirements through a two-pronged approach.

First, we provide a **minimal empirical demonstration** using a synthetic continual-adaptation setup. This experiment simulates cross-session model fine-tuning across multiple users and measures the degree to which identity separability increases when adaptive parameters persist across sessions. The purpose of this demonstration is not domain realism, but architectural observability: to show that cross-session adaptive persistence can measurably induce identity-correlated structure in model parameters under standard optimization procedures.

Second, we present a **domain-specific case study** in adaptive Brain–Computer Interface (BCI) systems—a deployment context characterized by high-resolution physiological coupling and iterative decoder adaptation. The BCI instantiation illustrates how NOIRÉA insertion points map to a high-risk adaptive pipeline where identity-correlated neural structure is empirically known to persist across sessions.

The synthetic experiment establishes empirical observability of Persistent Identity Modeling (PIM) as a structural phenomenon. The BCI case study illustrates architectural applicability in a physiologically coupled Physical AI domain. Together, these two validation tracks provide both measurement grounding and domain relevance.

F. Paper Organization

The remainder of this paper is organized as follows. Section II formalizes the threat model and defines PIM as a structural risk class. Section III specifies the NOIRÉA architectural requirements (R1–R4). Section IV presents the reference architecture and domain instantiation, including the BCI case study. Section V discusses compliance verification and validation methodology. Section VI positions NOIRÉA relative to related work in privacy-preserving machine learning, Physical AI safety, and human-computer interaction. Section VII discusses limitations and future extensions. Section VIII concludes.

II. STRUCTURAL MODEL AND PROBLEM FORMALIZATION

To formally characterize Persistent Identity Modeling (PIM) as a structural reliability risk, we first define a minimal adaptive system model.

A. System Model

Let the adaptive system state at session t be defined as:

$$M_t = M_t^{\text{task}} \oplus M_t^{\text{identity}} \quad (1)$$

where M_t^{task} denotes task-adaptive parameters optimizing performance, M_t^{identity} denotes identity-correlated representations derived from user-specific signals, and $\oplus$ denotes parameter space decomposition.

The system evolves according to:

$$M_{t+1} = f(M_t, H_t) \quad (2)$$

where H_t represents human-derived input signals at session t .

B. Session Boundary Specification

A **session** σ_i represents a temporal interval of continuous user interaction. At session initialization:

$$M_{t=0}^{\text{identity}} = \mathbf{0} \quad (3)$$

where $\mathbf{0}$ denotes zero initialization of the identity-correlated parameter partition, unless explicitly authorized by the user.

This defines the intended representation lifecycle boundary: identity-correlated state should not persist across session boundaries without authorization.

C. Cross-Session Persistence and PIM

Cross-session persistence occurs when:

$$M_{t+1}^{\text{identity}} \neq \mathbf{0} \quad (4)$$

after a session boundary transition without explicit user authorization.

Persistent Identity Modeling (PIM) is characterized by the following conditions:

- 1) **Cross-session state retention:** $M_{t+1}^{\text{identity}} \neq \mathbf{0}$ after session boundary
- 2) **Absence of explicit authorization:** No user-initiated renewal event
- 3) **Measurable behavioral influence:**

$$\Phi(M_t^{\text{identity}}) > \tau \quad (5)$$

where $\Phi(\cdot)$ denotes a measurable influence metric (e.g., identity separability via linear probing), and τ is a detectability threshold

Condition (3) ensures that persistence is not merely latent storage, but functionally influential on system behavior.

D. Risk Characterization

PIM introduces three structural risks:

- 1) **Identity Embedding Drift:** Model parameters progressively encode stable individual-specific features across sessions.
- 2) **Implicit Continuity Assumption:** The system assumes cross-session identity consistency without explicit user authorization.
- 3) **Boundary Opacity:** Users cannot distinguish between session-local adaptation and long-term representation persistence.

Importantly, PIM may arise as an unintended byproduct of performance optimization. The threat does not require adversarial intent; it emerges structurally from adaptive learning pipelines.

E. Distinction from Related Concerns

Unlike privacy violations (data leakage), behavioral misalignment (output deviation from specified values), and system malfunction (functional failure), PIM concerns **representation lifecycle integrity** at the architectural level—specifically, the unintended persistence of identity-correlated model state beyond declared session scope.

F. Assumptions and Scope

This work assumes: (1) adaptive models update parameters across sessions; (2) session boundaries exist but are not strictly enforced by default; (3) human-derived signals contain identity-correlated structure; and (4) no explicit mechanism exists to isolate or reset cross-session representation state.

We do not assume malicious data exfiltration. The threat considered here is structural accumulation inherent in adaptive Physical AI systems.

III. ARCHITECTURAL REQUIREMENTS

The structural decomposition of M_t in Section II isolates M_t^{identity} as a distinct representational component whose persistence introduces cross-session reliability risk. The following requirements (R1–R4) operationalize the architectural constraints necessary to bound, monitor, and control the evolution of M_t^{identity} over time. Together, these requirements translate the formal model into implementable structural conditions for representation lifecycle integrity.

A. R1 — Session-Bounded Identity Isolation

Requirement R1: At initialization of session σ_i , the identity-correlated state must satisfy:

$$M_{i,0}^{\text{identity}} = \mathbf{0} \quad (6)$$

unless explicitly authorized by the user.

No identity-correlated parameters from prior session σ_{i-1} may influence the new session by default.

Rationale: This enforces the session boundary condition defined in Section II and prevents unintended cross-session state carryover.

B. R2 — Controlled Cross-Session Persistence

Requirement R2: Any persistence of M^{identity} across sessions must be: (1) explicitly user-authorized, (2) logically separable from task-adaptive state, and (3) revocable without affecting M^{task} .

Formally, if persistence is enabled:

$$M_{i,0}^{\text{identity}} = P(M_{i-1,T}^{\text{identity}}) \quad (7)$$

where $P(\cdot)$ is an authorized persistence operator. Unauthorized persistence is prohibited.

Rationale: This requirement prevents structural accumulation of identity state through unconstrained adaptive updates.

C. R3 — Bounded Identity Influence

Let $\Phi(M_t^{\text{identity}})$ denote an influence metric quantifying the separability or recoverability of identity-specific information from the internal model state. In this work, Φ may be instantiated as a linear probe-based separability score such as ROC-AUC (Section V).

NOIRÉA requires:

$$\Phi(M_t^{\text{identity}}) \leq \tau \quad (8)$$

where τ is a predefined threshold reflecting acceptable identity persistence. The threshold τ is domain-dependent; a value

near chance level (e.g., $\tau \approx 0.55$ – 0.70 for ROC-AUC) would indicate that identity-correlated state remains below reliable detectability. Specific threshold selection requires domain-specific analysis.

Rationale: This ensures that identity persistence is not functionally opaque and remains within measurable bounds.

D. R4 — Revocable Identity State Reset

Requirement R4: Users must be able to revoke identity-correlated state independently of task-adaptive state. Formally, there must exist a reset operator $\mathcal{R}$ such that:

$$\mathcal{R}(M_t) = (M_t^{\text{task}}, \mathbf{0}) \quad (9)$$

without degrading baseline task performance beyond predefined tolerance ϵ .

Rationale: This enforces representation lifecycle integrity and prevents irreversible identity embedding drift.

E. Design Principles

NOIRÉA does not prohibit adaptation. Instead, it enforces session-scoped learning, identity-decoupled parameterization, and explicit representation lifecycle control. The objective is structural mitigation of unintended persistent identity modeling without degrading task performance.

IV. REFERENCE ARCHITECTURE AND DOMAIN INSTANTIATION

A. General Physical AI Reference Model

A general Physical AI system operating in sustained human interaction can be modeled as a sequential processing pipeline comprising: (1) *Sensor Layer*—acquires raw signals (motion, neural, physiological, tactile); (2) *Feature Extraction*—reduces raw signals to task-relevant representations; (3) *Adaptive Model*—updates internal parameters based on extracted features; and (4) *Execution Layer*—generates actions based on model output.

PIM may emerge at the *Adaptive Model* stage. When model parameters are updated iteratively across sessions without explicit state isolation, identity-correlated structure accumulates in the feature and parameter spaces. This accumulation is not visible at the Execution Layer and does not manifest as observable malfunction.

Session boundaries, if not architecturally enforced, permit cross-session parameter carry-over. The result is a model that implicitly encodes stable user-specific representations across deployment time.

B. NOIRÉA Boundary Insertion Points

NOIRÉA specifies three structural insertion points within the general pipeline (see Fig. 1):

Insertion Point 1 — Feature Isolation Boundary. Positioned between Feature Extraction and Adaptive Model. Extracted features must not retain identity-correlated structure beyond the current session scope. Session-scoped feature normalization or decorrelation must be applied to feature representations prior to adaptive model input.

TABLE I
MAPPING OF REQUIREMENTS TO INSERTION POINTS

Requirement	Insertion Point	Mechanism
R1 (Session-Bounded Isolation)	Session Boundary	Reset
R2 (Controlled Persistence)	Model Boundary	Parameter
R3 (Bounded Influence)	Feature + Model Boundary	$\Phi \leq \tau$ enforcement
R4 (Revocable Reset)	All boundaries	$\mathcal{R}(M_t)$ operator; logged events

NOIRÉA System Integration in Physical AI Pipeline

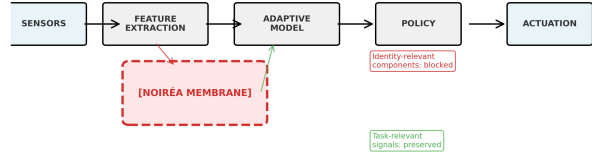


Fig. 1. NOIRÉA system integration in a Physical AI pipeline. The NOIRÉA membrane is positioned between Feature Extraction and Adaptive Model, filtering identity-correlated representations while preserving task-relevant signals. Identity-relevant components are blocked at the Adaptive Model boundary; task-relevant signals pass through to the Policy and Actuation layers.

Insertion Point 2 — Model Parameter Boundary. Positioned between Feature Extraction and Adaptive Model. Model parameter updates must be partitioned to prevent stable cross-session identity embedding. Parameters encoding task adaptation must be separable from parameters encoding identity-correlated state.

Insertion Point 3 — Session Reset Boundary. Positioned at session transition events. An explicit reset mechanism must be invoked at each session boundary. Post-reset model state must satisfy identity-subspace isolation constraints. Reset events must be logged as verifiable state transitions.

Table I maps insertion points to requirements.

C. Case Study: Adaptive Brain–Computer Interface Systems

We instantiate the NOIRÉA reference architecture in the domain of adaptive Brain–Computer Interface (BCI) systems [8]. This domain exhibits high-risk structural conditions for PIM: continuous neural signal acquisition, iterative decoder adaptation across sessions, and direct physiological coupling with the user.

1) *Standard Adaptive BCI Pipeline:* A representative adaptive BCI pipeline [9], [10] operates through neural signal acquisition (EEG/ECOG/spike trains), signal preprocessing (filtering, artifact removal), feature extraction (band power, CSP, neural embeddings), decoder adaptation (LDA, Riemannian methods, neural nets), and control output (cursor, prosthetic, communication).

Decoder adaptation typically employs supervised recalibration, unsupervised adaptation, or transfer learning. In the transfer learning and unsupervised cases, cross-session parameter persistence is explicit by design. In supervised recalibration, persistence may occur implicitly if session boundaries are not enforced at the parameter level.

2) *PIM Risk in Adaptive BCI*: Neural signals contain stable individual-specific structure. Prior work in EEG biometrics and neural signal analysis [11]–[13] has demonstrated that brain activity patterns exhibit individual-specific characteristics that persist across sessions. EEG and ECoG recordings encode anatomical, physiological, and behavioral features that distinguish users at the signal level. Decoder parameters trained on these signals may therefore accumulate identity-correlated embeddings across sessions.

Formally, if decoder parameters M_t are updated as $M_{t+1} = f(M_t, H_{t+1})$ where H_{t+1} contains identity-correlated neural features, then M_{t+1} may encode persistent identity structure derived from both H_{t+1} and prior sessions through M_t . This satisfies the PIM definition of Section II.

3) *NOIRÉA Insertion in Adaptive BCI*: Applying the reference architecture to the adaptive BCI pipeline:

Feature Isolation Boundary (between Preprocessing and Feature Extraction): Neural features must be session-scoped. Cross-session feature statistics (e.g., covariance matrices used in Riemannian adaptation) must be reset or re-estimated from session-local data only, unless explicit user authorization is present.

Model Parameter Boundary (between Feature Extraction and Decoder): Decoder parameters must be partitioned into task parameters (encoding control mapping) and identity parameters (encoding user-specific neural structure). Cross-session transfer is permitted only for task parameters. Identity-correlated parameter subspaces must satisfy R2 isolation requirements.

Session Reset Boundary (at session transition): Session boundaries must trigger explicit parameter state verification. Reset events must be logged with timestamps and parameter state checksums to satisfy R3 and R4.

4) *Compliance Verification in BCI Context*: A NOIRÉA-compliant adaptive BCI system must satisfy: (1) feature extraction outputs do not retain cross-session user-identifying statistics; (2) decoder parameter updates are partitioned and logged at session boundaries; (3) cross-session parameter carry-over is explicitly authorized and bounded; and (4) session boundary events are observable to supervisory control layers.

D. Implementation Constraints

NOIRÉA insertion points impose concrete dataflow and state management constraints on Physical AI system implementations.

1) *Parameter Partitioning Requirement*: Adaptive model parameters must be partitionable into disjoint subspaces:

$$M = M_{\text{task}} \oplus M_{\text{identity}} \quad (10)$$

Algorithm 1 Session Boundary Reset

Require: M_t (session t model state)

Ensure: M_{t+1} (reset model state)

- 1: Extract M_{identity} from M_t via parameter partition
 - 2: Reinitialize M_{identity} to baseline state ε
 - 3: Preserve M_{task} from M_t
 - 4: Construct $M_{t+1} = M_{\text{task}} \oplus \varepsilon$
 - 5: Compute checksum $h = \mathcal{H}(M_{t+1})$
 - 6: Log event: (timestamp, session_id, h)
 - 7: **return** M_{t+1}
-

where M_{task} encodes task-relevant adaptive state and M_{identity} encodes user-specific structure. Cross-session transfer is permitted only for M_{task} . Implementation must provide explicit parameter indexing or tagging such that M_{identity} components can be isolated and reset independently.

The decomposition in (10) represents a structural abstraction rather than a guarantee of strict orthogonality in practical implementations. In deep learning systems, task-relevant and identity-correlated parameters are often entangled. Practical realization may therefore rely on approximate subspace separation mechanisms, including layer isolation, adapter modules, low-rank parameter decomposition, or orthogonal projection constraints. The architectural requirement concerns controllable separability rather than exact mathematical independence.

2) *Session Boundary Reset Protocol*: At session transition $t \rightarrow t+1$, the system must execute the session boundary reset protocol described in Algorithm 1.

The baseline state ε may be a pretrained initialization, random initialization, or domain-specific neutral configuration, provided it does not encode identity-correlated structure from prior sessions.

3) *Identity-Subspace Isolation Constraint*: The reset mechanism must satisfy:

$$\text{corr}(M_{t+1}^{\text{identity}}, H_{\leq t}) < \tau_{\text{isolation}} \quad (11)$$

where $H_{\leq t}$ represents all human-derived signals from sessions up to t , and $\tau_{\text{isolation}}$ is a system-defined identity isolation threshold.

In practice, this constraint can be enforced through feature decorrelation (session-scoped whitening or PCA reset), parameter reinitialization (resetting identity-sensitive layers while preserving task-sensitive layers), or differential parameterization (separate parameter banks for cross-session and within-session adaptation).

4) *Logging and Observability Requirements*: Session boundary events must generate tamper-evident log records containing session identifiers, timestamps (ISO 8601 format, synchronized clock), parameter state checksums (cryptographic hash of M_t pre-reset and M_{t+1} post-reset), and authorization records (if cross-session transfer explicitly authorized). Logs must be accessible to supervisory control layers per R4, enabling external verification of session boundary enforcement without requiring access to raw model parameters.

V. COMPLIANCE VERIFICATION AND EMPIRICAL VALIDATION

A. Verification Approach

NOIRÉA requirements are designed to be architecturally verifiable. Compliance assessment can be conducted through standard architectural inspection and operational monitoring. Three complementary verification modalities support compliance assessment:

- 1) **Static architectural inspection:** verification that R1–R4 structural requirements are present in system design
- 2) **Log-based operational verification:** confirmation that session boundary events and reset mechanisms are executed as specified
- 3) **Behavioral compliance testing:** detection of identity-correlated state persistence through controlled multi-user scenarios

These modalities leverage existing verification practices in dependable systems engineering [17], [18].

B. Synthetic Demonstration of Representation Separability

To evaluate whether Persistent Identity Modeling produces empirically detectable representation divergence, we construct a controlled synthetic experiment with parametrized identity signal strength. The objective is not to approximate realistic continual learning dynamics, but to isolate the structural condition under which cross-session accumulation becomes statistically detectable—and to demonstrate that session boundary reset mitigates this accumulation.

1) *Experimental Design:* Each representation vector $x \in \mathbb{R}^d$ ($d = 50$) is sampled from a multivariate Gaussian $x \sim \mathcal{N}(\mu_i, I)$, where $\mu_i = \pm\alpha \cdot \mathbf{1}$ encodes a weak identity-specific offset and α modulates identity signal strength. Each session generates $N = 16$ samples (8 per identity class). The experiment simulates $K = 25$ deployment sessions across 30 independent runs with distinct random seeds.

Three conditions are compared:

- 1) **Baseline (unconstrained):** A logistic regression classifier is trained on accumulated data from all K sessions ($K \times N = 400$ samples) and evaluated on a held-out session. This simulates cross-session state persistence without reset.
- 2) **NOIRÉA (session-bounded):** The classifier is trained on a single session ($N = 16$ samples) and evaluated on a subsequent session. This simulates R1-compliant session reset.
- 3) **Shuffle control:** The classifier is trained on accumulated data with identity labels randomly permuted, verifying that baseline performance is attributable to identity-correlated structure rather than sample volume.

Identity separability is quantified as $\Phi = \text{ROC-AUC}$ on the held-out test session.

2) *Results:* Table II summarizes detection performance across identity signal strengths $\alpha \in \{0.0, 0.05, 0.10, 0.15, 0.20\}$; Fig. 2 presents the corresponding ROC curves and AUC bar charts.

TABLE II
IDENTITY SEPARABILITY (Φ , ROC-AUC) ACROSS CONDITIONS

α	Baseline	NOIRÉA	Shuffle
0.00	0.52 ± 0.14	0.51 ± 0.16	0.43 ± 0.16
0.05	0.67 ± 0.11	0.56 ± 0.16	0.44 ± 0.16
0.10	0.83 ± 0.09	0.66 ± 0.15	0.44 ± 0.16
0.15	0.92 ± 0.07	0.79 ± 0.13	0.44 ± 0.17
0.20	0.97 ± 0.04	0.89 ± 0.10	0.43 ± 0.17

At $\alpha = 0$, all conditions produce near-chance AUC, confirming absence of spurious detection. For $\alpha \geq 0.05$, baseline separability increases monotonically with identity signal strength, while NOIRÉA-constrained separability remains significantly lower (independent-samples t -test, $p < 0.002$ for all $\alpha > 0$). Shuffle control performance remains near chance across all conditions, confirming that baseline separability reflects identity-correlated structure rather than a sample volume artifact.

At $\alpha = 0.10$, the baseline condition ($\Phi = 0.83$) exceeds a reasonable detectability threshold, while the NOIRÉA condition ($\Phi = 0.66$) remains below it—illustrating how session reset prevents weak per-session signals from accumulating into detectable cross-session identity persistence.

3) *Interpretation:* The results demonstrate three properties relevant to NOIRÉA requirements. First, cross-session accumulation amplifies weak identity signals that are individually insufficient for reliable classification, producing measurable representation divergence (PIM observability). Second, session boundary reset per R1 significantly reduces identity separability by preventing cross-session signal accumulation. Third, residual within-session separability ($\Phi > 0.5$ in the NOIRÉA condition at higher α) reflects inherent identity structure in the input data, not cross-session persistence—consistent with the architectural scope of NOIRÉA, which constrains representation *lifecycle* rather than eliminating identity information from individual sessions.

This experiment constitutes a proof-of-observability for PIM under controlled synthetic conditions. Critically, the experiment demonstrates not merely that identity can be classified—which is well established in biometric literature—but that *weak signals individually insufficient for reliable classification become detectable through cross-session accumulation*, and that session boundary reset structurally prevents this amplification. This accumulation-driven observability is the architectural property that NOIRÉA is designed to constrain. Extension to realistic adaptive learning dynamics remains a direction for future work.

C. Illustrative Example: BCI Compliance

In the adaptive BCI context (Section IV), compliance can be tested analogously. User A completes multiple sessions; decoder state is recorded. Session boundary reset is invoked per R3. User B begins a new session with the post-reset system. Model behavior under User B is compared to baseline (fresh decoder) and post-User-A state.

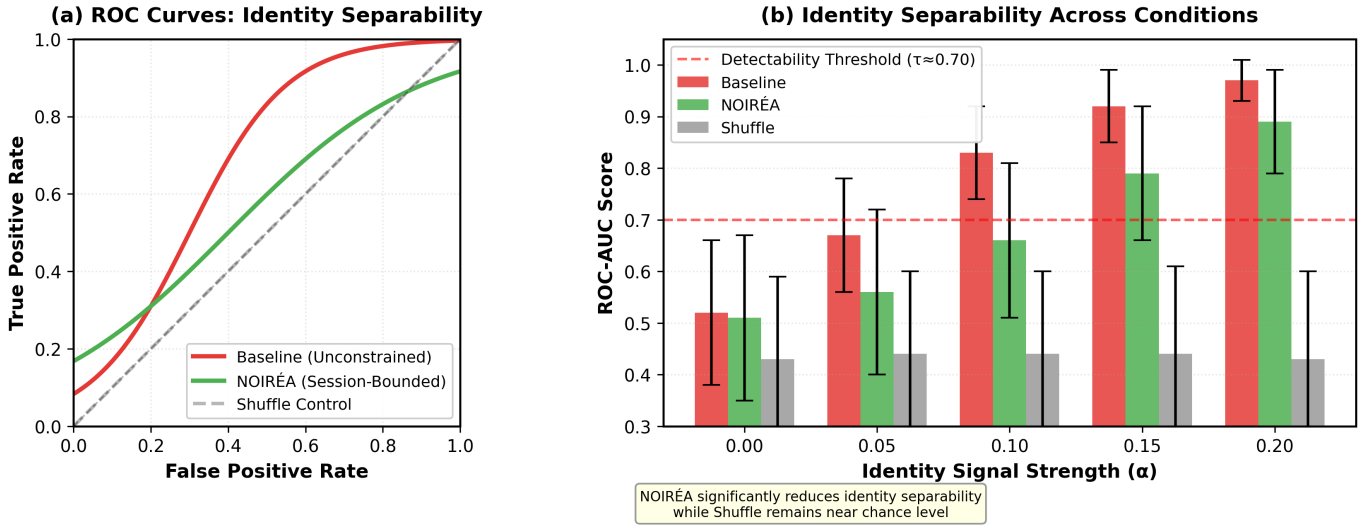


Fig. 2. Identity separability results across conditions and identity signal strengths. (a) ROC curves for Baseline (unconstrained), NOIRÉA (session-bounded), and Shuffle control conditions. (b) ROC-AUC scores across $\alpha \in \{0.0, 0.05, 0.10, 0.15, 0.20\}$. The detectability threshold $\tau \approx 0.70$ is shown as a dashed red line. NOIRÉA significantly reduces identity separability relative to Baseline ($p < 0.002$ for all $\alpha > 0$), while Shuffle remains near chance level across all conditions.

Compliance criterion: Model behavior under User B should not be statistically distinguishable from baseline with respect to User A-specific neural features. Persistent distinguishability indicates potential R2 violation (identity-correlated parameter retention).

Unlike the synthetic demonstration, this illustration reflects a physiologically grounded domain where neural signals are empirically known to exhibit stable individual structure [11]. The BCI case highlights how PIM risk may manifest in high-resolution human-machine coupling contexts.

D. Scope and Limitations

NOIRÉA verification addresses architectural conformance to R1–R4 within the defined threat model (Section II). It does not: guarantee absence of all forms of identity inference from system outputs; replace general privacy auditing or regulatory compliance assessment; or address threat models beyond cross-session representation persistence as formalized in this work.

Detailed verification methodology development, including formalization of identity-subspace projections and compliance thresholds, is outside the scope of this architectural specification and remains a direction for future work.

VI. RELATED WORK

A. Adaptive Physical AI Systems

Adaptive learning across deployment sessions is a standard design feature in Physical AI systems, including collaborative robotics, adaptive prosthetics, and Brain-Computer Interfaces [8], [9]. Cross-session model fine-tuning is widely employed to improve performance stability and reduce calibration time [10].

Prior work primarily evaluates such systems in terms of accuracy, convergence behavior, and robustness to noise. The

lifecycle of internal representations across sessions, however, is typically treated as an implementation detail rather than an architectural design constraint. This paper explicitly elevates cross-session representation persistence to an architectural concern.

B. Privacy-Preserving Machine Learning

Privacy-preserving learning techniques such as differential privacy [3], [4], federated learning [5], and data minimization frameworks provide statistical guarantees against specific inference attacks. These methods aim to prevent reconstruction of sensitive data from model outputs or gradients.

However, such approaches do not explicitly constrain the structural persistence of identity-correlated representations within adaptive model parameters across sessions. A system may satisfy differential privacy guarantees while still accumulating stable cross-session identity embeddings at the architectural level.

NOIRÉA addresses representation lifecycle isolation rather than statistical privacy guarantees.

C. Functional Safety and Dependability Standards

Functional safety standards commonly applied to safety-critical Physical AI deployments (e.g., IEC 61508 [1], ISO 26262 [2]) focus on preventing malfunction, loss of control, and hazardous operational states. These frameworks specify fault tolerance, redundancy, and fail-safe design principles.

They do not, however, define architectural constraints governing cross-session representation persistence in adaptive learning pipelines. NOIRÉA complements these standards by introducing structural requirements at the representation lifecycle level. Dependable systems research [17], [18] provides foundational terminology and risk taxonomy that informs the

characterization of PIM as a reliability-relevant structural risk class.

D. User Modeling and Continual Learning

Continual learning and personalization frameworks [6], [7] intentionally leverage cross-session state persistence to improve user-specific adaptation. In many contexts, such persistence is desirable and functionally beneficial.

In contrast, NOIRÉA addresses deployment contexts where cross-session identity embedding constitutes a structural risk. Rather than eliminating adaptation, it constrains representation persistence through session-bounded architectural isolation.

VII. DISCUSSION

A. Scope and Applicability

NOIRÉA addresses architectural constraints on cross-session representation persistence in adaptive Physical AI systems. The requirements apply to deployment contexts where systems operate in sustained interaction with individual human users across multiple sessions, adaptive learning mechanisms update model parameters based on human-derived signals, and session boundaries exist as temporal or operational divisions in deployment lifecycle.

NOIRÉA does not address short-term interactions, stateless systems, or purely informational AI tools where cross-session adaptation does not occur.

B. Assumptions and Prerequisites

The architectural requirements assume: (1) *identifiable session boundaries*—systems can delineate temporal or operational session transitions; (2) *parameter partitionability*—model architectures admit separation of task-relevant and identity-correlated parameter subspaces; and (3) *supervisory control access*—session boundary events and model state are observable to authorized control layers.

These assumptions reflect design choices rather than inherent limitations. Systems not currently satisfying these properties may require architectural refactoring to achieve NOIRÉA compliance.

C. Design Trade-offs

NOIRÉA enforcement introduces architectural complexity and potential performance considerations.

Complexity. Explicit session boundary management, parameter partitioning, and logging infrastructure increase system implementation overhead compared to unconstrained adaptive pipelines.

Parameter decoupling challenge. The requirement for $M = M_{\text{task}} \oplus M_{\text{identity}}$ represents an idealized decomposition. In deep learning models such as neural decoders, task-relevant and identity-correlated parameters are often entangled. Practical implementations may achieve approximate decoupling through orthogonal projection, adversarial training, or layer-wise parameter isolation, though perfect separation remains an open challenge.

Adaptation efficiency. Session-bounded learning may reduce cross-session transfer benefits in scenarios where user-specific adaptation genuinely improves task performance. The trade-off between identity isolation and adaptation efficiency is deployment-context dependent.

Implementation burden. Retrofitting existing adaptive systems to satisfy R1–R4 may require non-trivial architectural changes, particularly in systems where cross-session state management was not originally designed with isolation constraints.

These trade-offs reflect engineering decisions rather than fundamental barriers. In deployment contexts where cross-session identity accumulation constitutes a structural risk, the architectural overhead is justified by the mitigation objective.

D. Limitations

This work does not provide formal verification methods (while R1–R4 are designed to be verifiable, formal methods for proving identity-subspace isolation remain future work); define universal identity-subspace projections (identity-correlated features are domain-specific); address all forms of identity inference (NOIRÉA constrains architectural representation persistence but does not eliminate all potential inference paths); or specify performance guarantees (the impact of R1–R4 on task-level adaptation performance is system-dependent).

E. Generalization Beyond BCI

The BCI case study serves as an illustrative high-risk instantiation. While the architectural principles are domain-agnostic, practical instantiation requires domain-specific analysis. In collaborative robotics, identity-correlated features may include motion preferences, interaction timing, or force application patterns. In adaptive exoskeletons, gait signatures, muscle activation patterns, or joint angle preferences may encode user-specific structure. In assistive devices, behavioral patterns, control strategies, or environmental interaction characteristics may persist across sessions.

Each domain requires specification of what constitutes identity-correlated state within that context. NOIRÉA provides the architectural framework; domain experts must define the identity subspace.

F. Relationship to Existing Frameworks

NOIRÉA is designed to complement, not replace, existing safety and governance frameworks. Functional safety standards [1], [2] address malfunction and loss of control. Privacy frameworks (GDPR [16], differential privacy [3]) address data protection and statistical privacy guarantees. AI alignment approaches [14], [15] address value specification and behavioral correctness.

Integration with these frameworks is a direction for future work, particularly in regulated deployment contexts such as medical devices or safety-critical assistive systems.

G. Reliability Positioning

From a reliability perspective, Persistent Identity Modeling is conceptualized not as a runtime fault, but as a structural hazard arising from unconstrained adaptive state persistence across sessions.

Unlike transient faults or performance degradation, PIM does not manifest as immediate malfunction. Instead, it introduces latent state coupling between temporally separated deployments, potentially violating independence assumptions commonly implicit in dependable system design [17]. The system-level consequence is not operational failure in the conventional sense, but a structural compromise of the conditions under which reliability claims are valid: testing protocols that assume session independence may produce misleading coverage results, and certification arguments that rely on behavioral equivalence across deployment epochs may be invalidated by hidden identity-correlated state carry-over.

By characterizing cross-session representation persistence as a distinct reliability-relevant risk class in adaptive Physical AI systems, this work reframes representation lifecycle management as a core concern in dependable system design. Rather than treating adaptive state accumulation as an implementation detail, NOIRÉA positions representation persistence within the scope of architectural risk analysis, providing a foundation for future extensions of reliability theory to long-term adaptive systems.

VIII. CONCLUSION

This paper introduced NOIRÉA, a structural constraint architecture that addresses Persistent Identity Modeling (PIM) in adaptive Physical AI systems.

We formalized PIM as a structural risk class arising from a specification gap in current safety and governance frameworks: existing approaches address malfunction, privacy breaches, and alignment failures, but do not specify architectural requirements for cross-session representation lifecycle management. NOIRÉA fills this gap through four architectural requirements (R1–R4) that enforce session-bounded isolation of identity-correlated model state.

The reference architecture demonstrates that these requirements can be instantiated as concrete insertion points within adaptive learning pipelines, without requiring novel verification techniques or implementation replacement. The adaptive BCI case study illustrates how NOIRÉA constraints apply to high-risk Physical AI contexts characterized by sustained physiological coupling and iterative decoder adaptation.

Beyond BCIs, the architectural principles may generalize to other adaptive Physical AI contexts where sustained human interaction creates potential for cross-session identity accumulation. Collaborative robotics, adaptive exoskeletons, and assistive devices similarly exhibit iterative learning over human-derived signals, though domain-specific instantiation of R1–R4 would require additional analysis. As such systems transition from laboratory prototypes to long-term deployment, representation lifecycle management represents an important consideration in dependable system design.

Future work includes formalization of identity-subspace projection operators, development of automated compliance verification tools, and empirical evaluation of NOIRÉA-compliant architectures across multiple deployment contexts. The relationship between session-boundary constraints and task performance under varying adaptation regimes remains an open question requiring controlled empirical study.

NOIRÉA contributes to dependable system design by characterizing cross-session identity accumulation as a structural risk class rather than an implementation detail. By formally identifying and bounding this risk at the architectural level, this work establishes representation lifecycle integrity as an architectural concern in the design of adaptive Physical AI systems that operate in sustained interaction with human users.

ACKNOWLEDGMENTS

The author acknowledges AI language model assistance in formalizing technical content and refining manuscript structure, while retaining full responsibility for conceptual contributions, research decisions, and experimental validation.

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